Module-1
INTRODUCTION TO SENSOR-BASED MEASUREMENT SYSTEMS

1.1 GENERAL CONCEPTS AND TERMINOLOGY
1.1.1 Measurement Systems
A system is a combination of two or more elements, subsystems, and parts necessary to carry out one or more functions. The function of a measurement system is the objective and empirical assignment of a number to a property or quality of an object or event in order to describe it. That is, the result of a measurement must be independent of the observer (objective) and experimentally based (empirical). Numerical quantities must fulfill the same relations fulfilled by the described properties. For example, if a given object has a property larger than the same property in another object, the numerical result when measuring the first object must exceed that when measuring the second object. One objective of a measurement can be process monitoring: for example, ambient temperature measurement, gas and water volume measurement, and clinical monitoring. Another objective can be process control: for example, for temperature or level control in a tank.
Figure 1.1 shows the functions and data flow of a measurement and control system. In general, in addition to the acquisition of information carried out by a sensor, a measurement requires the processing of that information and the presentation of the result in order to make it perceptible to human senses. Any of these functions can be local or remote, but remote functions require information transmission. Modern measurement systems are not physically arranged according to the data flow in Figure 1.1 but are instead arranged according to their connection to the digital bus communicating different subsystems.

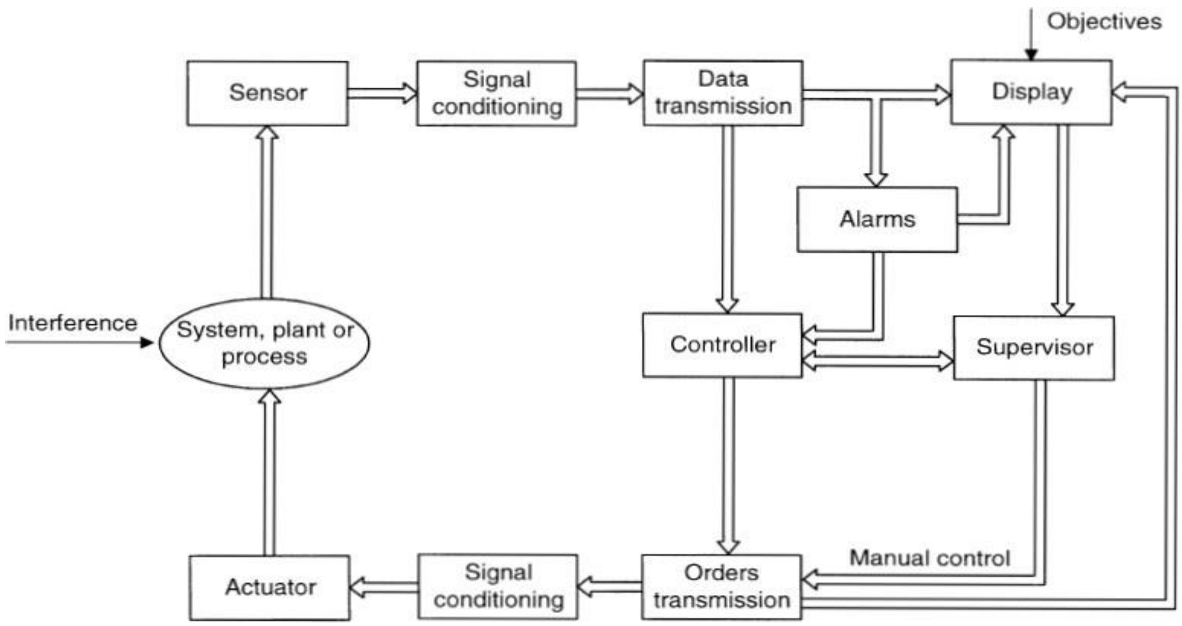

Figure 1.1 Functions and data flow in a measurement and control system. Sensors and actuators are transducers at the physical interface between electronic systems and processes or experiments.

1.1.2 Transducers, Sensors, and Actuators
A **Transducer** is a device that converts a signal from one physical form to a corresponding signal having a different physical form. Therefore, it is an energy converter. This means that the input signal always has energy or power; that is, signals consist of two component quantities whose product has energy or power dimension.

Since there are six different kinds of signals - mechanical, thermal, magnetic, electric, chemical, and radiation (corpuscular and electromagnetic, including light) - any device converting signals of one kind to signals of a different kind is a transducer. The resulting signals can be of any useful physical form. Devices offering an electric output are called **Sensors.** Most measurement systems use electric signals and hence rely on sensors. Electronic measurement systems provide the following advantages:

1. Sensors can be designed for any nonelectric quantity, by selecting an appropriate material. Any variation in a nonelectric parameter implies a variation in an electric parameter because of the electronic structure of matter.
2. Energy does not need to be drained from the process being measured because sensor output signals can be amplified. Electronic amplifiers yield (low) power gains exceeding 10^{10} in a single stage. The energy of the amplifier output comes from its power supply. The amplifier input signal only controls (modulates) that energy.
3. There is a variety of integrated circuits available for electric signal conditioning or modification. Some sensors integrate these conditioners in a single package.
4. Many options exist for information display or recording by electronic means. These permit us to handle numerical data and text, graphics, and diagrams.
5. Signal transmission is more versatile for electric signals. Mechanical, hydraulic, or pneumatic signals may be appropriate in some circumstances, such as in environments where ionizing radiation or explosive atmospheres are present, but electric signals prevail.

Actuators are designed mainly for power conversion. Sometimes, particularly when measuring mechanical quantities, a primary sensor converts the measurand into a measuring signal. Then a sensor would convert that signal into an electric signal. For example, a diaphragm is a primary sensor that stresses when subject to a pressure difference, and strain gages sense that stress.

1.1.3 Signal Conditioning and Display

Signal conditioners are measuring system elements that start with an electric sensor output signal and then yield a signal suitable for transmission, display, or recording, or that better meet the requirements of a subsequent standard equipment or device. They normally consist of electronic circuits performing any of the following functions: amplification, level shifting, filtering, impedance matching, modulation, and demodulation. Some standards call the sensor plus signal conditioner subsystem a transmitter. One of the stages of measuring systems is usually digital and the sensor output is analog. Analog-to-digital converters (ADCs) yield a digital code from an analog signal. ADCs have relatively low input impedance, and they require their input signal to be dc or slowly varying, with amplitude within specified margins, usually less than ±10 V. Therefore, sensor output signals, which may have an amplitude in the millivolt range, must be conditioned before they can be applied to the ADC. The display of measured results can be in an analog (optical, acoustic, or tactile) or in a digital (optical) form. The recording can be magnetic, electronic, or on paper, but the information to be recorded should always be in electrical form.

1.1.4 Interfaces, Data Domains, and Conversion

In measurement systems, the functions of signal sensing, conditioning, processing, and display are not always divided into physically distinct elements. Furthermore, the border between signal conditioning and processing may be indistinct. But generally, there is a need for some signal processing of the sensor output signal before its end use. Some authors use the term interface to refer to signal-modifying elements that operate in the electrical domain, even when changing from one data domain to another, such as an ADC. A data domain is the name of a quantity used to represent or transmit information. The concept of data domains and conversion between domains helps in describing sensors and electronic circuits associated with them.

Figure 1.2 shows some possible domains, most of which are electrical. In the analog domain the information is carried by signal amplitude (i.e., charge, voltage, current, or power). In the time domain the information is not carried by amplitude but by time relations (period or frequency, pulse width, or phase). In the digital domain, signals have only two values. The information can be carried by the number of pulses or by a coded serial or parallel word. The analog domain is the most prone to electrical interference. In the time domain, the coded variable cannot be measured - that is, converted to the numerical domain - in a continuous way. Rather, a cycle or pulse duration must elapse. In the digital domain, numbers are easily displayed. The structure of a measurement system can be described then in terms of domain conversions and changes, depending on the direct or indirect nature of the measurement method. Direct physical measurements yield quantitative information about a physical object or action by direct comparison with a reference quantity. This comparison is sometimes simply mechanical, as in a weighing scale. In indirect physical measurements the quantity of interest is calculated by applying an equation that describes the law relating other quantities measured with a device, usually an electric one. For example, one measures the mechanical power transmitted by a shaft by multiplying the measured torque and speed of rotation, the electric resistance by dividing dc voltage by current, or the traveled distance by integrating the speed. Many measurements are indirect.

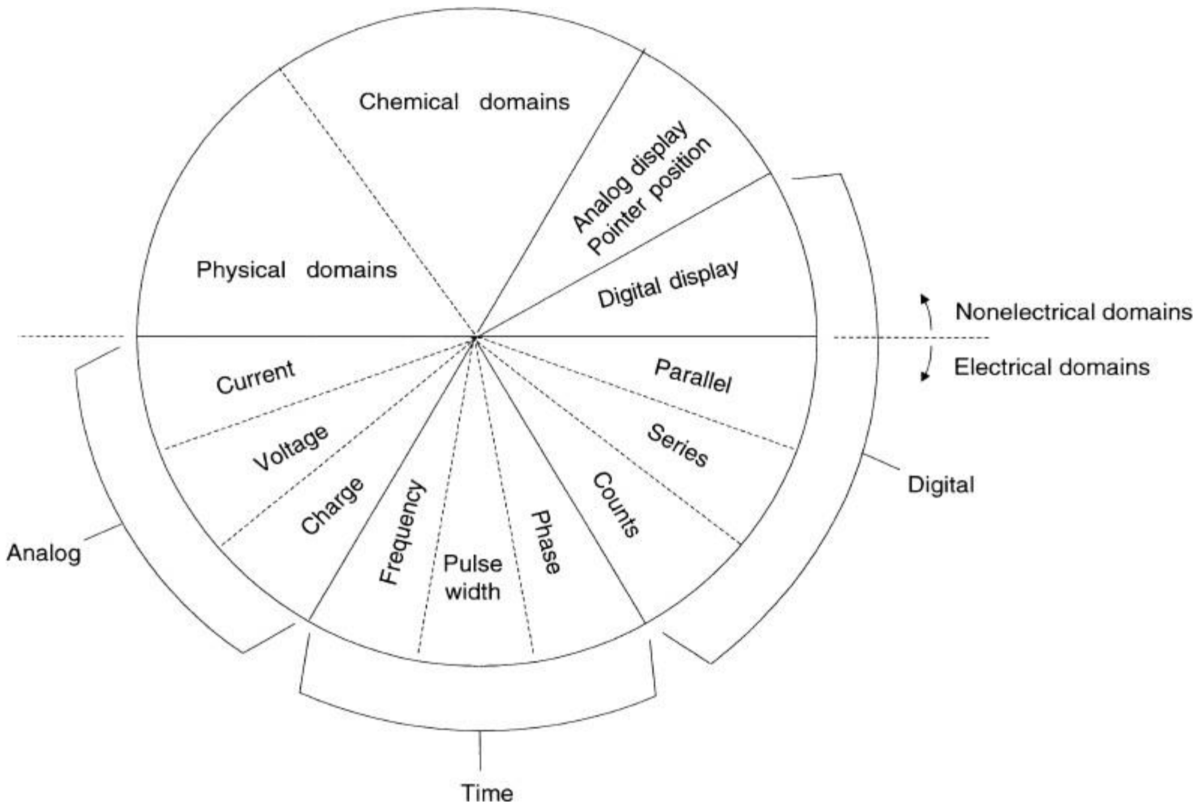

Figure 1.2 Data domains are quantities used to represent or transmit information.

1.2 SENSOR CLASSIFICATION

A great number of sensors are available for different physical quantities. In order to study them, it is advisable first to classify sensors according to some criterion.

In considering the need for a power supply, sensors are classified as modulating or self-generating. In modulating (or active) sensors, most of the output signal power comes from an auxiliary power source. The input only controls the output. Conversely, in self-generating (or passive) sensors, output power comes from the input. Modulating sensors usually require more wires than self-generating sensors, because wires different from the signal wires supply power.

Moreover, the presence of an auxiliary power source can increase the danger of explosion in explosive atmospheres. Modulating sensors have the advantage that the power supply voltage can modify their overall sensitivity. Some authors use the terms active for self-generating and passive for modulating. To avoid confusion, we will not use these terms. In considering output signals, we classify sensors as analog or digital. In analog sensors the output changes in a continuous way at a macroscopic level.
The information is usually obtained from the amplitude, although sensors with output in the time domain are usually considered analog. Sensors whose output is a variable frequency are called quasi-digital because it is very easy to obtain a digital output from them (by counting for a time). The output of digital sensors takes the form of discrete steps or states. Digital sensors do not require an ADC, and their output is easier to transmit than that of analog sensors. Digital output is also more repeatable and reliable and often more accurate. But regrettably, digital sensors cannot measure many physical quantities.
In considering the operating mode, sensors are classified in terms of their function in a detection or a null mode. In detection sensors the measured quantity produces a physical effect that generates in some part of the instrument a similar but opposing effect that is related to some useful variable. For example, a dynamometer to measure force is a sensor where the force to be measured detects a spring to the point where the force it exerts, proportional to its deformation, balances the applied force. Null-type sensors attempt to prevent detection from the null point by applying a known effect that opposes that produced by the quantity being measured. There is an imbalance detector and some means to restore balance. In a weighing scale, for example, the placement of a mass on a pan produces an imbalance indicated by a pointer. The user has to place one or more calibrated weights on the other pan until a balance is reached, which can be observed from the pointer's position. Null measurements are usually more accurate because the opposing known effect can be calibrated against a high-precision standard or a reference quantity.
Table 1.1 compares the classification criteria above and gives examples for each type in different measurement situations.

TABLE 1.1 Sensor Classifications According to Different Exhaustive Criteria

Criterion	Classes	Examples
Power supply	Modulating	Thermistor
	Self-generating	Thermocouple
Output signal	Analog	Potentiometer
	Digital	Position encoder
Operation mode	Deflection	Deflection accelerometer
	Null	Servo-accelerometer

1.3 PRIMARY SENSORS

Primary sensors convert measurands from physical quantities to other forms.
We classify primary sensors here according to the measurand. Devices that have direct electric output are plain sensors.

1.3.1 Temperature Sensors: Bimetals

A bimetal consists of two welded metal strips having different thermal expansion coefficients that are exposed to the same temperature. As temperature changes, the strip warps according to a uniform circular arc (Figure 1.3).

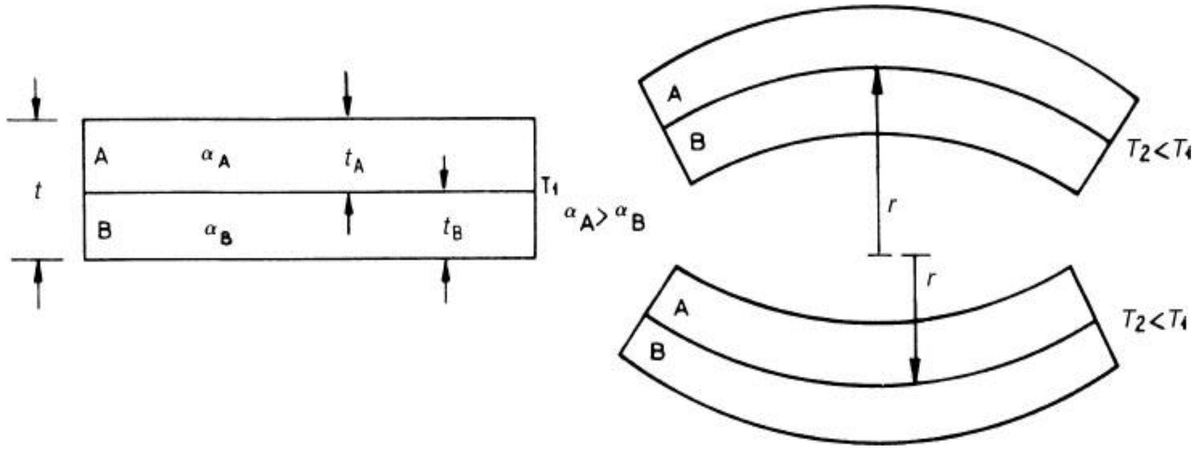

Figure 1.3 A bimetal consists of two metals with dissimilar thermal expansion coefficients, which deforms when temperature changes. Dimensions and curvature have been exaggerated to better illustrate the working principle.

If the metals have similar moduli of elasticity and thicknesses, the radius of curvature r, when changing from temperature T_1 to T_2, is

$$r \cong \frac{2t}{3(\alpha_A - \alpha_B)(T_2 - T_1)} \quad ....1.1$$

where t is the total thickness of the piece and where α_A and α_B are the respective thermal expansion coefficients. Therefore, the radius of curvature is inversely proportional to the temperature difference. A position or displacement sensor would yield a corresponding electric signal. Alternatively, the force exerted by a total or partially bonded or clamped element can be measured. The thickness of common bimetal strips ranges from 10 mm to 3 mm. A metal having $\alpha_B < 0$ would yield a small r, hence high sensitivity. Because useful metals have $\alpha_B > 0$, bimetal strips combine a high-coefficient metal (proprietary iron-nickel-chrome alloys) with invar (steel and nickel alloy) that shows $\alpha = 1.7 \times 10^{-6}$/°C. Micromachined actuators (microvalves) use silicon and aluminum. Bimetal strips are used in the range from -75°C to +540°C, and mostly from 0°C to +300°C. They are manufactured in the form of cantilever, spiral, helix, diaphragm, and so on, normally with a pointer fastened to one end of the strip, which indicates temperature on a dial. Bimetal strips are also used as actuators to directly open or close contacts (thermostats, on-off controls, starters for fluorescent lamps) and for overcurrent protection in electric circuits: The current along the bimetal heats it by Joule effect until reaching a temperature high enough to exert a mechanical force on a trigger device that opens the circuit and interrupts the current. Other non-measurement applications of bimetal strips are the thermal compensation of temperature-sensitive devices and fire alarms. Their response is slow because of their large mass.

1.3.2 Pressure Sensors

Pressure measurement in liquids or gases is common, particularly in process control and in electronic engine control. Blood pressure measurement is very common for patient diagnosis and monitoring. Pressure is defined as the force per unit area. Differential pressure is the difference in pressure between two measurement points. Gage pressure is measured relative to ambient temperature. Absolute pressure is measured relative to a perfect vacuum. To measure a pressure, it is either compared with a known force or its effect on an elastic element is measured (detection measurement).

A liquid-column manometer such as the U-tube in Figure 1.4a compares the pressure to be measured with a reference pressure and yields a difference h of liquid level. When second order effects are disregarded, the result is

$$h = \frac{p - p_{\text{ref}}}{\rho g} \qquad \text{...........1.2}$$

where r is the density of the liquid and g is the acceleration of gravity. A level sensor (photoelectric, float, etc.) yields an electric output signal. Elastic elements deform under pressure until the internal stress balances the applied pressure. The material and its geometry determine the amplitude of the resulting displacement or deformation.
Usual pressure sensors use the Bourdon tube, diaphragms, capsules, and bellows.
The Bourdon tube-patented by Eugene Bourdon in 1849 - is a curved (Figure 1.4b) or twisted (Figure 1.4c), flattened metallic tube with one closed end. The tube is obtained by deforming a tube having a circular cross section. When pressure is applied through the open end, the tube tends to straighten. The displacement of the free end indicates the pressure applied. This displacement is not linear along its entire range but is linear enough in short ranges. Displacement sensors yield an electric output signal. Tube configurations with greater displacements (spiral, helical) have large compliance and length that result in a small-frequency passband. The tube metal (brass, monel, steel) is selected to be compatible with the medium.
A diaphragm is a flexible circular plate consisting of a taut membrane or a clamped sheet that strains under the action of the pressure difference to be measured (Figure 1.4d).
The sensor detects the detection of the center of the diaphragm, its global deformation, or the local strain (by strain gages). Some metals used are beryllium-copper, stainless steel, and nickel-copper alloys. A micromachined diaphragm is an etched silicon wafer with diffused or implanted gages that sense local strain (Figure 1.4e). Cars and hospitals use silicon pressure sensors by the millions. The diaphragm and elements bonded on it must be compatible with the medium and withstand the required temperature. For a thin plate with thickness t and radius R experiencing a pressure difference Δp across it, if the center detection is z < t/3, we have

$$z \cong \frac{3(1 - \nu^2)R^4}{16Et^3}\Delta p \qquad \text{...........1.3}$$

where E is Young's modulus and υ the Poisson's ratio for the plate material.

Large, flexible diaphragms undergo large detection but have large compliance. Thin plates yield large detections but are fragile. An alternative to sense the central detection is to use a rod to transmit force to a cantilever beam with bonded strain gages, away from media temperature. Ceramic and some silicon pressure sensors rely on the capacitance change between an electrode applied on the diaphragm and one fixed electrode. Piezoresistive sensors distributed on the diaphragm can sense radial and tangential strain. They are connected in a measurement bridge to add their signal and compensate temperature interference. Capsules and bellows yield larger displacements than diaphragms. A capsule (Figure 1.4f) consists of twin corrugated diaphragms joined by their external border and placed on opposite sides of the same chamber. A bellows (Figure 1.4g) is a flexible chamber with axial elongation that undergoes detections larger than capsules, up to 10% of its length. Capsules and bellows are vibration and acceleration-sensitive, do not withstand high overpressures, and have high compliance, hence poor dynamic response.

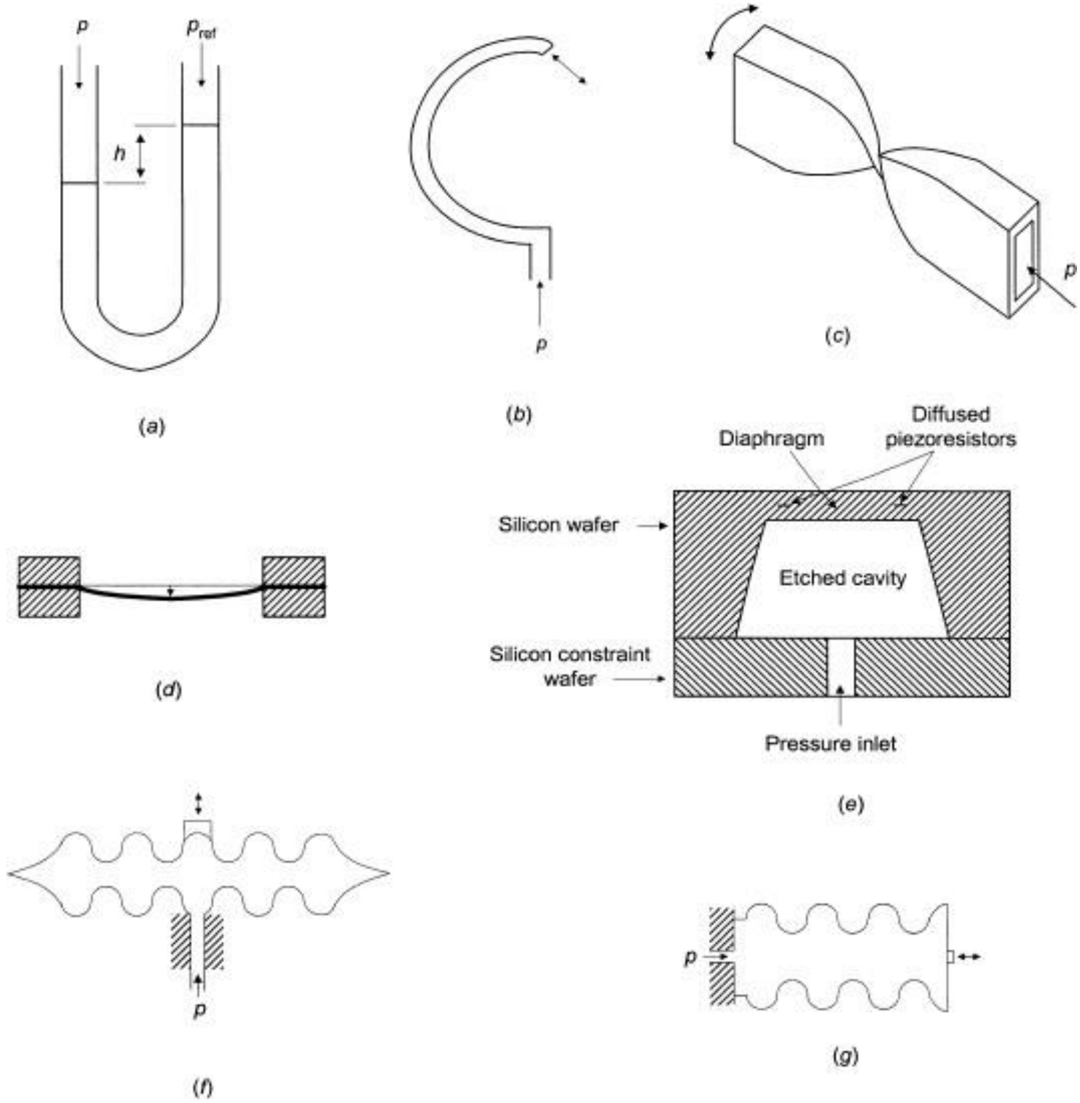

Figure 1.4 Primary pressure sensors. (a) Liquid-column U-tube manometer. The liquid must be compatible with the fluid for which pressure is to be measured, and the tube must withstand the mechanical stress. (b) C-shaped Bourdon tube. (c) Twisted Bourdon tube. (d) Membrane diaphragm. (e) Micromachined diaphragm. (f) Capsule. (g) Bellows. The area of the diaphragm in (e) is less than 1 mm^2. All other devices can measure up to several centimeters.

1.3.3 Level Sensors

Dipsticks are simple level sensors but cannot easily provide an electric signal. Floats, based on Archimedes' buoyancy principle, convert liquid level to force or displacement (Figures 1.5a and 1.5b). In sealed or high-pressure containers, the position of the float can be detected magnetically. Build-up and deposits on the float surface limit performance. The pressure of liquid or solid is proportional to level (Figure 1.5c), according to

$$h = \frac{\Delta p}{\rho g} \qquad \text{...........1.4}$$

where ρ is density and g is the acceleration of gravity. This method is suitable for both pressurized and open containers. Temperature interferes because it varies density.

The bubble tube in Figure 1.5d overcomes the need for a pressure port near the container bottom, which is a potential leak source. The dip tube has an open end close to the bottom of the tank. An inert gas flows through the dip tube and when gas bubbles escape from the open end, the gas pressure in the tube equals the hydraulic pressure from the liquid.

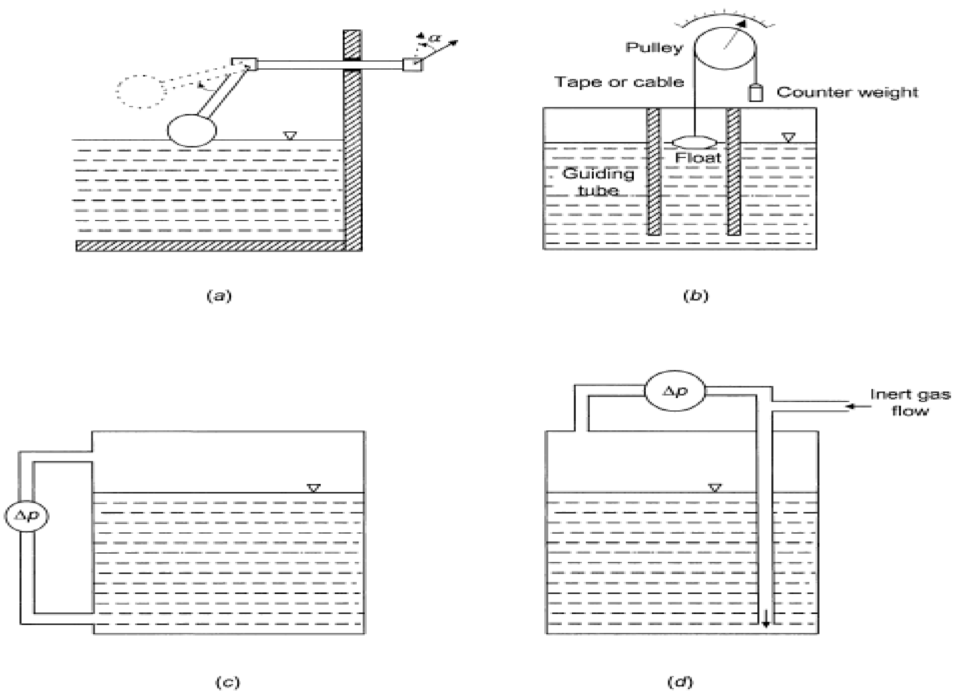

Figure 1.5 Primary level sensors. (a) and (b) Based on a float. (c) and (d) Based on differential pressure measurement.

1.4 MATERIALS FOR SENSORS

Sensors rely on physical or chemical phenomena and materials where those phenomena appear usefully - that is, with high sensitivity, repeatability and specificity. Those phenomena may concern the material itself or its geometry, and most of them have been known for a long time. Major changes in sensors come from new materials, new fabrication techniques, or both. Solids, liquids, and gases consist of atoms, molecules, or ions-atoms or group of atoms that have lost or gained one or more electrons. Atoms consist of a positive nucleus and electrons orbiting around it in shells. If the outer electron shell is not full, atoms try to gain extra electrons and become bonded in the process, forming molecules or agglomerates. There are four main bond types: ionic, metallic, covalent, and van der Waals. Ionic bonds result from the electrostatic attraction between ions of different polarity. Ionic bonds form crystals-solids whose atoms are arranged in a long-range three-dimensional pattern in a way that reduces the overall energy and maintains electrical neutrality. Ionic crystals, such as NaCl and CsCl, have low electrical conductivity (because there are no free charges), relatively high fusion temperature, and good mechanical resistance, all resulting from the strong cohesion between ions.

The relative separation between energy bands determines the electrical conductivity of materials, which is a useful property for sensors. Figure 1.6 shows that valence and conduction bands overlap in conductors, so that there are always free electrons and the electrical conductivity is high. Insulators have distant valence and conduction bands, and errant vibrations of electrons around their positions (e.g., because of thermal agitation) are unable to furnish enough carriers to significantly conduct electricity. Semiconductors have a narrower forbidden band than do insulators, and electrons excited by thermal, electric, optical or other energy form can jump across that band and contribute to conduct electricity. Each energy form has different efficacy on liberating electrons.

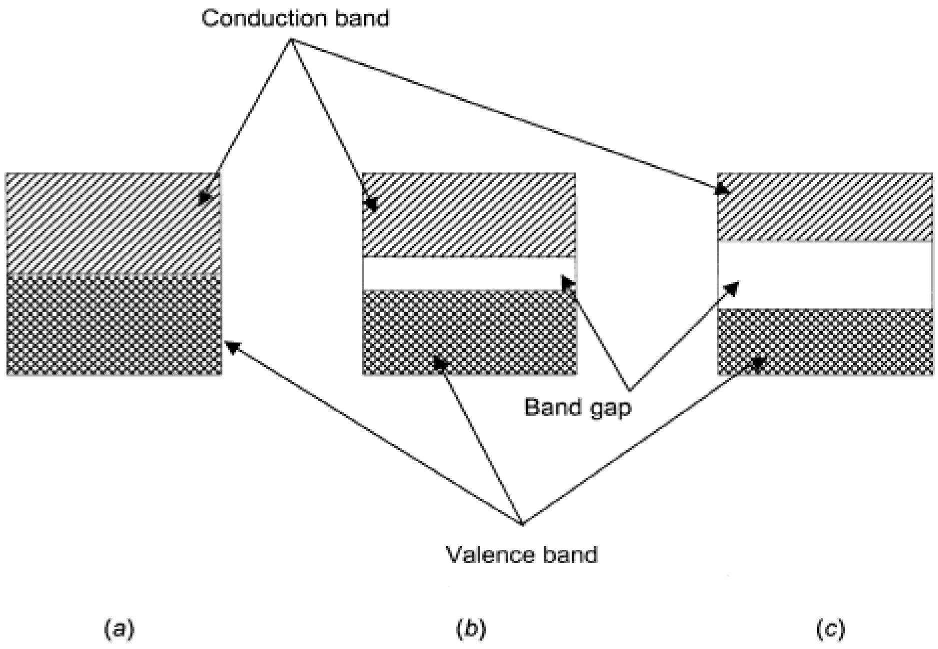

Figure 1.6 Energy bands for (a) a conductor, (b) a semiconductor, and (c) an electrical insulator.

According to their structure, solids can be single crystal, polycrystalline, amorphous, and glasses. Crystals can be considered the periodic repetition of a “unit cell" totally filling space. Unit cells can be described by a space lattice of points (Figure 1.7a). Atoms can occupy not only the vertices but also the center of the cell, the center of two or more faces, or combinations of them. Directions and lattice planes are referred to by the so-called Miller indices.

The direction of a vector is specified by the magnitude of its three components along the three axes, usually by writing them side by side enclosed in square brackets. A minus sign above a figure denotes a negative number (Figure 1.7b). Planes are specified by similarly writing inside parentheses the reciprocals of the intercepts of the plane on the axes of the unit cell (Figures 1.7c and 1.7d).

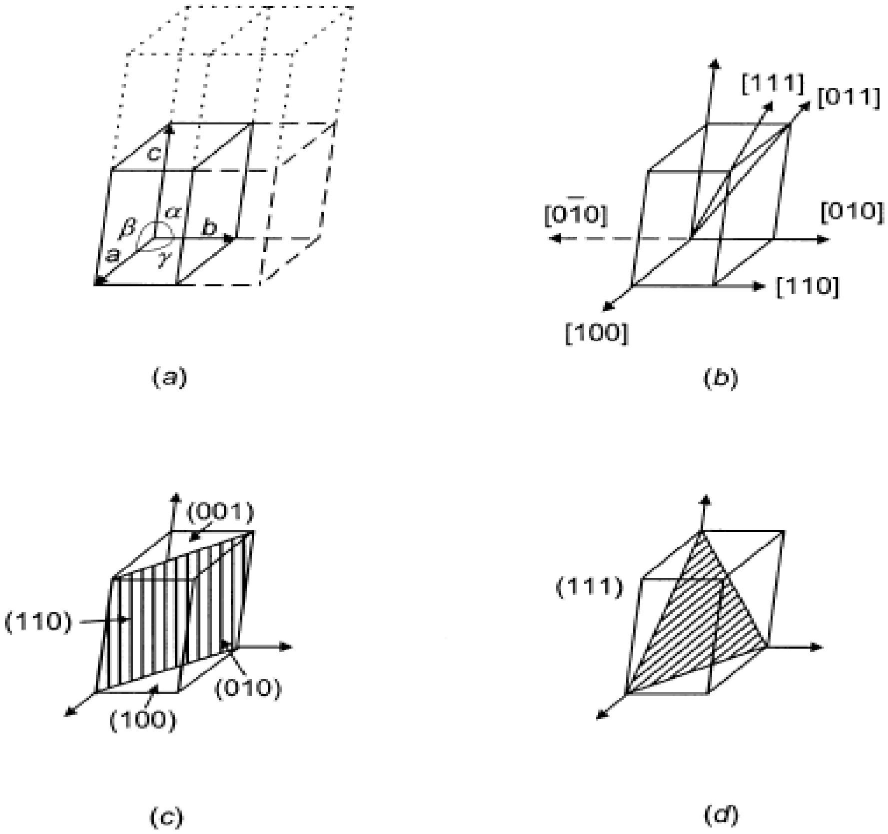

Figure 1.7 (a) Space lattice generated from a unit cell defined by vectors a, b, and c and angles a, b, and g between them. Miller indices describe the direction of a vector by giving the magnitudes of its three components along the three direction axes (b), and they describe planes by giving the reciprocals of their intercepts on the axes of the unit cell (c) and (d).

1.4.1 Conductors, Semiconductors, and Dielectrics

There are two type of conductors: electronic conductors (metals and their alloys) and ionic conductors or electrolytes (acid, base, or salt solutions). A potential difference V applied between the two ends of a solid of length l creates an internal electric field E = V/l, which accelerates electrons at

$$a = \frac{qE}{m} = \frac{qV}{ml}$$

where m is the electron mass and q its charge. Hence, electrons moving at random because of thermal agitation have in addition a velocity component in the direction of the applied field.
However, collisions deviate electrons from the direction set by E. It can be shown that the average or drift velocity is $v_d = a\tau$, where $a\tau$ is the average time between collisions, mean free time, relaxation time, or collision time. Therefore,

$$v_d = \frac{qE}{m}\tau = \frac{q\tau}{m}E$$

where $q\tau/m$ is termed the electron mobility μ_e.

The current density crossing a unit area will be

$$J = N_e q v_d = \frac{N_e q^2 \tau}{m} E$$

where N_e is the density of electrons (number of electrons per unit volume). This is Ohm's law and

$$\sigma = \frac{N_e q^2 \tau}{m} = \mu_e (N_e q)$$

is the electrical conductivity. A high conductivity can result from a high mobility or a high density of electrons. Because of random atom vibration, electron mobility in metals is relatively small. They are good conductors because of the abundance of free electrons. Metals and their alloys are used in sensors because of (a) their thermoelectric properties and (b) the dependence of their electrical conductivity with temperature and stress; they are also used to form electric circuits in which a measurand produces significant changes. Primary sensors such as bimetals and elastic elements (e.g., diaphragms and load cells) also rely on metals and their alloys. Still other metals are used because of their magnetic properties, or they are used as electrodes or to catalyze. Electrolytes are primarily used in chemical sensors.
Semiconductors are extensively used in sensors. Semiconductors have covalent bonds, hence low electrical conductivity. Because both electrons and holes contribute to the total current, transforms into

$$\sigma = q(N_n \mu_n + N_p \mu_p) = q(n\mu_n + p\mu_p)$$

where $n = N_n$ and $p = N_p$ are the respective concentrations of free electrons and holes. This conductivity depends on temperature, stress, electric and magnetic fields, corpuscular and electromagnetic radiation (including light), and the absorption of different substances.

1.4.2 Magnetic Materials

The magnetic flux in vacuum is proportional to the applied magnetic field,

$$B = \mu_0 H$$

where $\mu_0 = 4\pi \times 10^{-7}$ H/m is the permeability of vacuum. All materials modify the magnetic flux to some extent, so that

$$B = \mu_0 (H + M) = \mu_0 \mu_r H$$

where M is the magnetic dipole moment per unit volume, or magnetization, and μ_r is the relative permeability.
The magnetic properties of solids are related to the properties of the electrons in their atoms. Paramagnetic materials. $\mu_r > 1$. have atoms or ions with incomplete electron shells. Unpaired individual electron spins yield a magnetic moment, but the random orientation of individual dipoles makes the net dipole moment negligible. Nevertheless, upon application of an external field, individual dipoles assume the direction of minimal energy, setting themselves parallel to the applied field. Hence, applied magnetic fields attract paramagnetic materials.

Thermal agitation opposes that alignment, so that only at absolute zero (0 K) would the alignment be perfect (Curie-Weiss law). Diamagnetic materials. $\mu_r < 1$. have atoms or ions with complete electron shells; hence they lack magnetic moment. However, an applied magnetic field imparts an additional rotation to electrons (Larmor precession). This electron movement yields a net magnetic moment in a direction opposed to the field. Hence, applied magnetic fields repel diamagnetic materials. That alignment is not influenced by temperature.

Elementary magnetic dipoles in ferromagnetic materials arise from elementary currents. For iron and metals in its group (cobalt, nickel), those currents come from unpaired electron spins.

In rare earth elements, such as gadolinium, orbital unbalance also contributes elementary currents.

A ferromagnetic material magnetizes because of two different processes: displacement and orientation. Displacement refers to the volume change of some domains at the expense of their neighbors. Orientation refers to the alignment of the magnetic domains in the direction of the applied magnetic field. Figure 1.8 describes the magnetization process for an increasing applied field H. When H is small (Figure 1.8a), domains parallel to it or with close direction grow, and the material becomes slightly magnetized in that direction. But upon reducing H the domains resume their initial sizes and the magnetization disappears. Hence, for weak external fields, magnetization is reversible.

As H increases, the induced magnetization increases almost proportionally (Figure 1.8b) because of the reorientation of elementary domains. Nevertheless, after reducing H, some domains shrink and other rotate. As a result, the remanence-remaining magnetization-is somewhat smaller than that under the applied field. A strong enough H aligns all magnetic domains (Figure 1.8c) so that further field increases do not cause a larger magnetization. The material becomes saturated. If the external field decreases to zero, some domains forced beyond their stable state rotate and the remanence decreases below that corresponding to saturation. The opposing magnetic intensity that should be applied to remove the residual magnetism is termed coercive force or field.

As the magnetic force cycles from positive to negative values, the material describes a specific B-H curve different for increasing and decreasing H. Magnetic materials are classified according to the relative size of the hysteresis curve. Soft magnetic materials (Figure 1.8d) have a narrow hysteresis cycle with large μ_r (usually larger than 1000) and have a coercive force smaller than 100 μT. They suit ac applications because their magnetization can be easily reversed. Examples are silicon-iron, ferroxcube, permalloy, and mu-metal. Hard magnetic materials follow a wide hysteresis cycle (Figure 1.8e) with relatively small magnetic permeability and large coercive force (more than 100 mT in general), which makes them attractive for permanent magnets. They are also mechanically hard, hence difficult to work. Examples are carbon steel, alnico V(Al-Ni-Co) and hycomax (Al-Ni-Co-Cu).

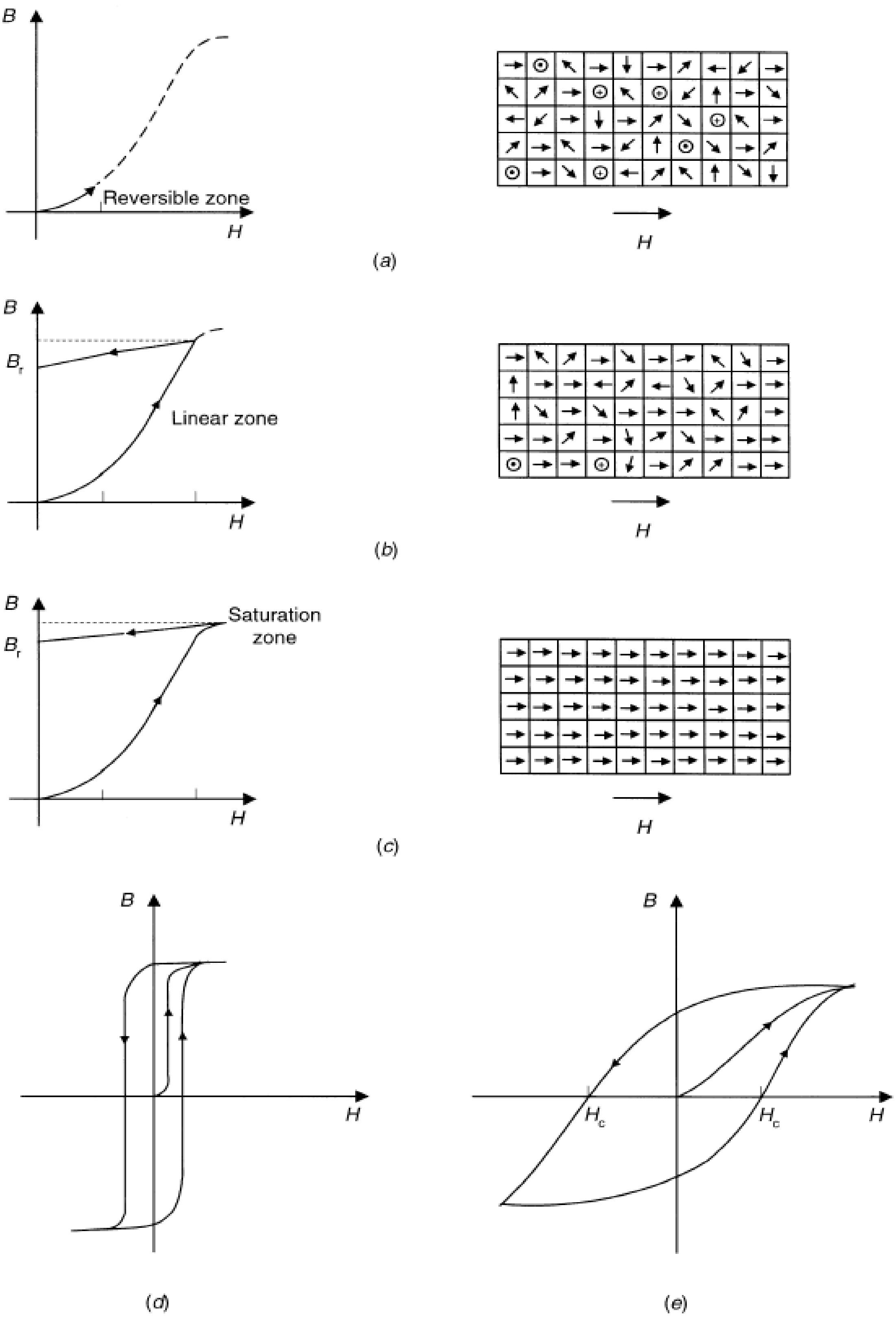

Figure 1.8 The magnetization of ferromagnetic materials results from domain displacement and orientation. (a) The magnetization under weak magnetic force is reversible. (b) The magnetization for stronger magnetic force is almost proportional to it, but the remanence is smaller than the induction under the applied field. (c) An intense external field saturates the material. (d) Soft magnetic materials have a narrow hysteresis cycle. (e) Hard magnetic materials have a wide hysteresis cycle.

1.5 MICROSENSOR TECHNOLOGY

Microsensor materials are prepared according to their nature, the desired sensing principle, and the intended application. There is an increasing interest in applying integrated circuit (IC) technology and micromachining, because they yield small, reliable sensors produced in large amounts leading to low cost.

1.5.1 Thick-Film Technology

Thick-film technology uses pastes or ``inks'' with fine particles (5 mm in average diameter) of common or noble metals dispersed in an organic vehicle, along with a glass frit that binds them. Depending on the dispersed particles, the paste can be conductive, resistive, or dielectric. Those pastes are screen-printed on a substrate according to a predefined pattern involving width lines from 10 mm to 200 mm. The printed film is dried by heating at about 150 °C to remove the organic solvent that provided the low viscosity needed for the paste to squeeze through the open areas in the screen. The substrate with the deposited film is then fired on a conveyor belt furnace, usually in air atmosphere, so that the metal powder sinters and the glass frit melts, thereby bonding the film to the substrate. The printing, drying, and firing sequence is repeated for each paste used according to predetermined thermal cycles. The result is a 10 mm to 25 mm thick film, impermeable to many substances but relatively porous for specific chemical or biological agents. Thick-film components have a printed tolerance from G10% to G20 %, but they can be later trimmed to within G0.2% to G0.5% through selective abrasion or laser vaporization. Depending on the firing temperature, there are three basic thick-film circuit types. Low-temperature pastes melt below 250 °C and are deposited on plastic materials, including those for printed circuits (glass fiber with epoxy), or anodized aluminum. There are thermoplastic and thermoset pastes. Thermoplastic pastes are mostly used in membrane switches. Thermoset pastes are epoxy materials with carbon and silver. High-temperature pastes melt at 800 °C to 1000 °C and use alumina, sapphire, or beryl (a silicate of beryllium and aluminum). Conductive pastes embed palladium, ruthenium, gold, and silver. Dielectric pastes use borosilicate glass. Medium-temperature pastes are similar to high-temperature pastes, melt at about 500 °C to 650 °C, and are deposited on low-carbon steel with porcelain enamel.

Thick-film technology finds at least three different uses in sensors. It has been used for years to fabricate hybrid circuits (multichip modules) offering improved performance compared to monolithic integrated circuits for signal conditioning and processing. Thick-film circuits and some sensors can be integrated in the same package, which improves reliability (strong connections), permits functional trimming, and reduces cost. It is also used to create support structures onto which a sensing material is deposited. Some thick-film pastes directly respond to physical and chemical quantities. F. H. Nicoll and B. Kazan described the fabrication of a screen-printable photoconductor in 1955. There are pastes-some developed for sensing applications-with high temperature coefficient of resistance useful for temperature sensing, piezoresistive pastes, magnetoresistive pastes, photoconductive pastes, piezoelectric pastes, and pastes with high Seebeck coefficient, among others. Pastes based on organic polymers and metal oxides such as SnO_2 can detect humidity and gases because of adsorption and absorption. Using thick-film technology, it is straightforward to define the interdigitated structures required for those sensors. Thick-film sensors with ceramic substrate withstand high temperatures, can be driven with relatively large voltages and currents, can integrate heaters, and can resist corrosion. Because the paste is fired into the ceramic, thick-film sensors are compact and sturdy. The printing process is quite inexpensive, which permits competitive low volume fabrication.

1.5.2 Thin-Film Technology

Thin films are obtained by vacuum deposition on a substrate of polished, high purity (99.6 %) alumina or low-alkalinity glass. Sensor and circuit patterns are defined by masks and transferred by photolitography, similarly to monolithic IC fabrication. Even though their names may suggest that the only difference between thick-film and thin-film technology is in film thickness, they are quite different technologies. In fact, metallized thin films may become thicker than some ``thick" films.

Common materials in thin-film circuits are nichrome for resistors, gold for conductors, and silicon dioxide for dielectrics. Many thin-film sensors are resistive. Piezoresistors use nichrome and polycrystalline silicon, temperature and electrodes for conductivity sensors use platinum, anisotropic magnetoresistors use nickel, cobalt, and iron alloys, gas sensors use zinc oxide, and conductivity sensors use platinum.

Thin films are deposited by the same techniques used in IC fabrication: spin casting, evaporation, sputtering, reactive growth, chemical vapor deposition, and plasma deposition. In spin casting, the thin-film material is dissolved in a volatile solvent and poured on the fast rotating substrate. Upon rotation, the liquid spreads and the solvent evaporates, thereby leaving a solid, uniform layer 0.1 mm to 50 mm thick. Thin films can also form by evaporating the material in a vacuum chamber in the presence of the substrate. The thin-film source is held in a heated crucible, and its evaporated atoms condense on the substrate. Sputtering or cathodic deposition also uses a vacuum chamber but the film material is placed on an anode, evaporated by bombardment with plasma of an inert gas inside the chamber, and deposited on a substrate placed on a cathode. Materials able to react with the substrate can be deposited by allowing their reaction. This is termed reactive growth, extensively used to grow silicon dioxide from a silicon surface held at high temperature. In chemical vapor deposition (CVD) there is a heated chamber with gas inlets and outlets. The high temperature breaks down incoming gases (pyrolysis) that contain the atomic components of the film, and the resulting components impinge on the substrate where they nucleate forming a film. The outlet carries away reaction and exhaust gases. CVD films can conform to the substrate. Epitaxial growth is a special CVD process that yields monocrystalline films on crystalline substrates. This is the preferred process to manufacture diaphragms for pressure sensors. CVD can be enhanced by the presence of plasma (PECVD) that induces the decomposition of gaseous compounds into reactive species at lower temperature. Langmuir-Blodget films are ultrathin films named in honor of Irving Langmuir and Katherine Blodgett, who developed the technique in the 1920s.

These are monolayer films of usually insulating materials whose molecules have hydrophilic ``head" and hydrophobic ``tail." Upon dispersion on the surface of a trough of water (like oil in water), the head orients toward the water while the tail orients out of the water. If hydrophilic and hydrophobic forces are balanced, the result is a monolayer. These films can be transferred from the water by controlled dipping and deposited to form membranes for enzyme immobilization or to attract gas molecules. Monolayers can be transferred one by one and stacked. Langmuir-Blodgett films have been used in ISFETs and SAW sensors.

1.5.3 Micromachining Technologies

Micromachining refers to processes to obtain three-dimensional devices whose feature dimensions and spacing between parts are in the range of 1 mm or less. Because it is a batch process (i.e., a process performed on an entire wafer 200 mm in diameter yielding hundreds of devices) and because materials and processes are borrowed from proven IC technology, micromachining has dramatically improved the performance-to-cost ratio over that of conventionally machined sensors. Reduced size and mass have broadened the dynamic range for some mechanical measurands. Placing integral electronics in the sensor housing increases reliability, but at the cost of reduced operating temperature. Micromachined sensors and other semiconductor-based sensors are termed microsensors. The development of microsensors is cost effective for applications requiring several million sensors per year, such as those in the automotive, home appliances, and biomedical industries. Three-dimensional devices are manufactured in planar technology by using layers of coordinated two-dimensional patterns. The basic planar fabrication processes are deposition, lithography, and etching. Thin-film deposition has been described above. Photolithography is the process of defining a pattern on a layer by first covering that layer with a thin film of photoresist, then exposing it to a light through a quartz mask, followed by chemical development (Figure 1.9). The result is a pattern that leaves parts of the original layer exposed to undergo chemical etching, ion implantation, or other processing. Removing the photoresist leaves the layer with the transferred pattern. Photolithography is used to either (a) define the geometry of overlying thin films such as silicon dioxide, polycrystalline silicon, and silicon nitride, deposited on the substrate or (b) directly modify the properties of the substrate (silicon or other). Chemical etching relies on the oxidation of silicon to form compounds that can be removed from the substrate. Chemical etchants can be liquid, vapor, or plasma. Wet etching uses aqueous solutions of strong acids or bases that remove silicon from exposed areas of the uppermost layer from an immersed sample with a patterned photoresist. Some wet etches are isotropic; that is, the etching rate is the same for all directions. This creates undercuts that make the patterned structure somewhat smaller than the resist mask (Figure 1.10a). Other wet etches are anisotropic because certain silicon planes have different chemical reactivity (Figure 1.10b)-reactions are fast in the [100] and [110] directions and slow in the [111] direction. Quartz resists etching along planes parallel to its z-axis. Etching rate and direction can be controlled by adding dopants to the exposed areas: n-type silicon is removed 50 times faster than p-type silicon. Vapor-phase etchants are halogen molecules or compounds that dissociate into reactive halogen species upon adsorption by silicon, thus forming volatile silicon compounds. Vapor phase etches are isotropic. Plasma phase etching relies on high reactivity of free halogen radicals created in ionized plasma. Bombarding the ions perpendicularly to the silicon surface enhances the reaction and yields anisotropic etching (Figure 1.10c). Microstructures for sensors are mostly formed by bulk and surface micromachining, the former being more common. Bulk micromachining removes significant amounts of material from a relatively thick substrate, usually a single crystal of silicon, but also amorphous glass, quartz (crystalline glass), and gallium arsenide.

The etching process can be isotropic (Figure 1.10a) or anisotropic (Figures 1.10b and 1.10c). Some sensors use two or more wafers that are bonded together. Bulk micromachining is extensively used in pressure sensors and has also been applied in cantilever beam accelerometers. Surface micromachining builds three-dimensional structures from stacked and patterned thin films such as polysilicon, silicon dioxide, and silicon nitride.

Figure 1.11 illustrates some process steps. First the silicon substrate is thermally oxidized to give a SiO2 layer, onto which a silicon nitride mask (for protection or electrical insulation) is deposited. Then a sacrificial (or space) layer such as 2 mm thick phosphosilicate glass (PSG) is sputtered and patterned, followed by deposition of the structural layer (often polysilicon 1mm to 4mm thick). Upon removal of the sacrificial layer by selective etching, the microstructure stands free and electric contacts are added. Surface micromachining yields smaller devices than bulk micromachining. Some commercial accelerometers are polysilicon surface microstructures with integrated MOS electronics.

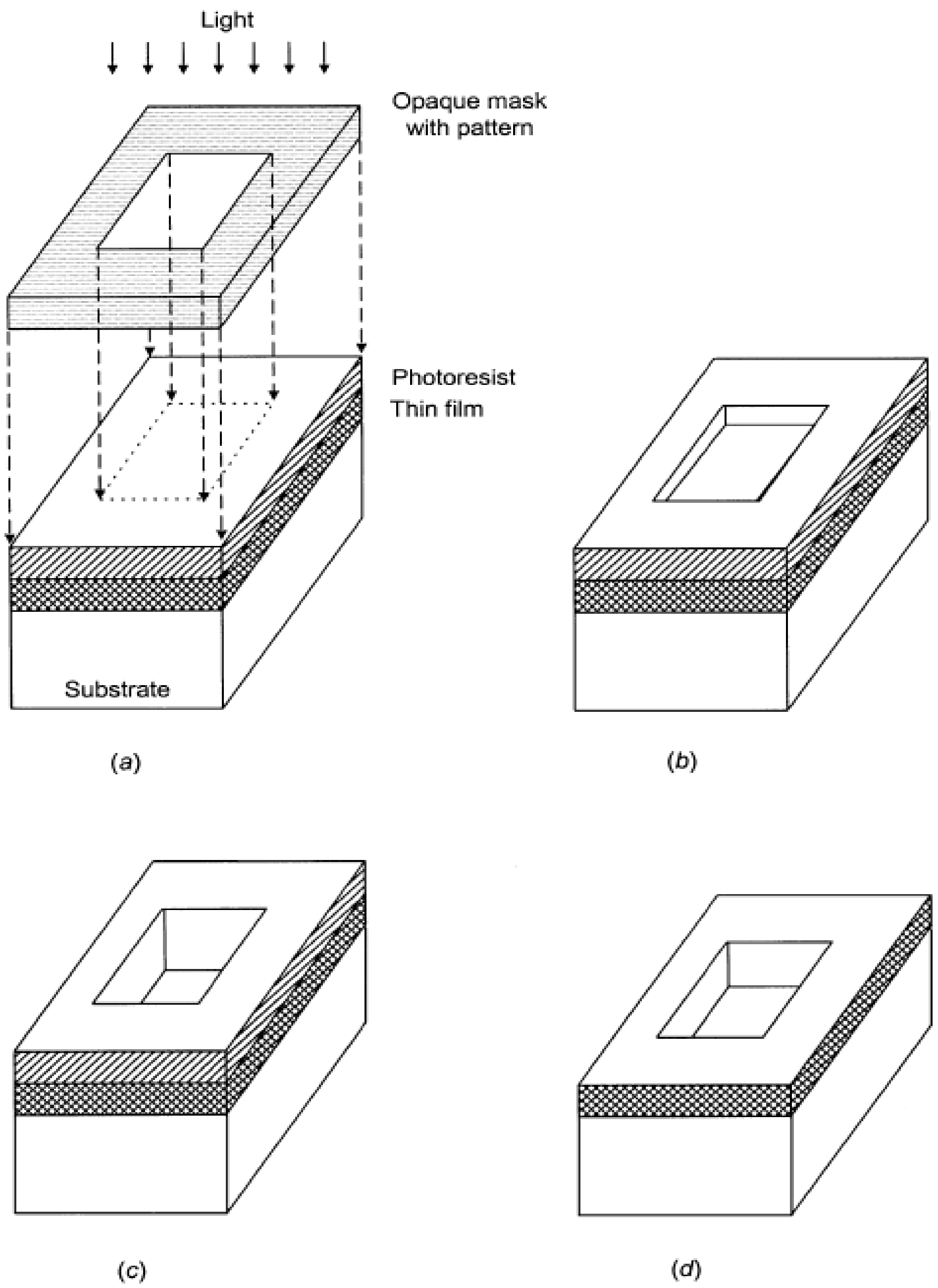

Figure 1.9 Planar photolithography processes. (a) The pattern is projected from a quartz mask onto photoresist. (b) Chemical development removes unsensitized photoresist. (c) Chemical etching removes the film uncovered by photoresist. (d) Photoresist removal leaves the desired pattern on film.

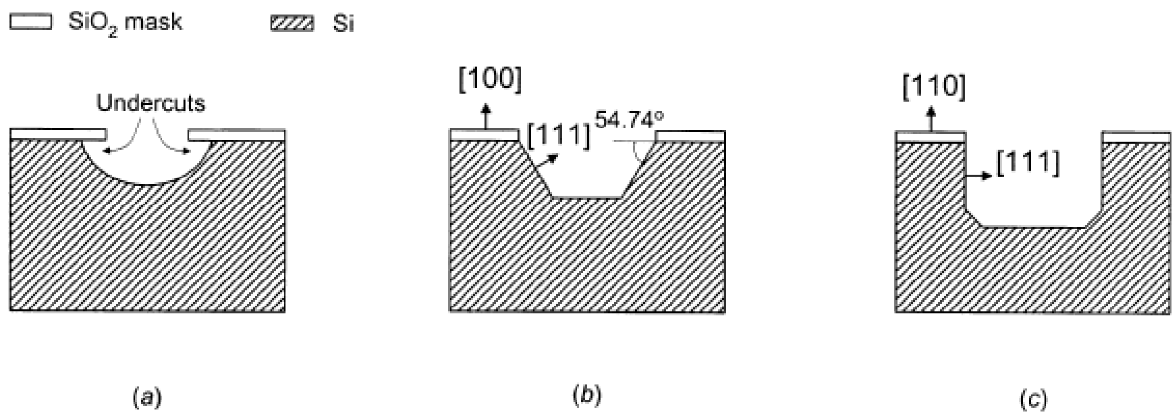

Figure 1.10 Etching can be (a) isotropic or (b, c) anisotropic, depending on the etchants and the etching method.

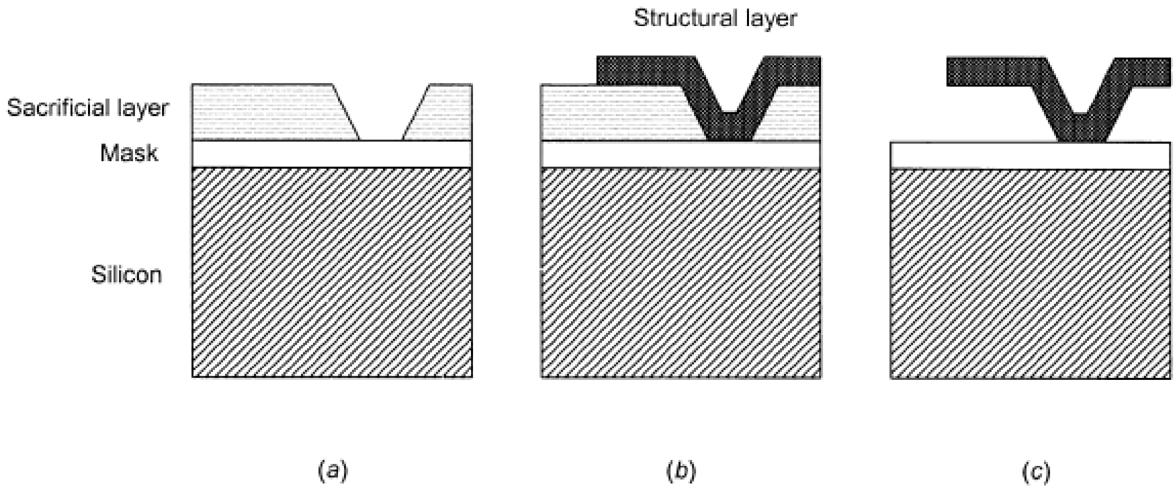

Figure 1.11 Surface micromachining processes include depositing (a) a sacrificial layer followed by (b) a body or structural layer, which becomes free upon removal of the sacrificial layer by (c) selective etching.